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Air stripping pre-treatment process to enhance biogas production in anaerobic digestion of chicken manure wastewater

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ABSTRACT

Anaerobic digestion is typically applied for water and air pollution control and energy recovery from chicken manure wastewater. Unfortunately, high nitrogen content of the chicken manure could be prohibitive to efficient anaerobic digestion. In this study, air stripping was employed as a pre-treatment process for ammonia removal from a chicken manure wastewater. Laboratory-scale experiments of air striping were conducted at various temperatures, initial pH values, air flow rates, and stripping times. The experimental results showed that the high ammonia removal efficiencies of more than 80% could be achieved from the air stripping process operating at the initial pH, temperature, and air flow rate of 11, 80 °C, and 4.0 L/min, respectively. The ammonia removal kinetics were found to fit with the first-order reaction. Bio-methane potential tests suggested that air stripping pre-treatment of the chicken manure wastewater resulted in about 25% increase in biogas production.

1. Introduction

Anaerobic digestion (AD) is a cost-effective and environmentally-friendly method which can treat and recover bio-energy from various types of organic wastes. AD of livestock wastes such as chicken and swine manures could yield considerable environmental and economic benefits, including water and air pollution control, and renewable energy production (Blumenstein et al., 2016; Cowley and Brorsen, 2018; Scarlat et al., 2018). Although chicken manure is rich in organic matters and has higher potentials for biogas production than swine and cow manures (Owamah et al., 2014; Qiao et al., 2011), its high nitrogen content could be prohibitive to efficient anaerobic digestion. During microbial degradation, organic nitrogen is converted to ammonia which could act as a strong inhibitor on microbiological conversion in the AD process (Angelidaki and Ahring, 1993; Nielsen and Angelidaki, 2008). Total ammonia nitrogen concentration greater than 3 g/L could result in a decreased methane production (Bi et al., 2020; Calli et al., 2005; Chen et al., 2008; Yenigün and Demirel, 2013). To overcome the ammonia inhibition of AD, various ammonia reduction techniques have been developed including: substrate dilution, co-digestion with high-carbon substrates, pH and temperature setting, use of additives with adsorptive capacity such as activated carbon, bentonite and zeolite, use of trace elements such as selenium, microflora acclimation and bioaugmentation, stripping, membrane processes, struvite precipitation, ultrasonication and microwave irradiation, and biological process such as Anammox (Fuchs et al., 2018).

Ammonia stripping is considered to be a cost-effective technique that requires relatively low energy inputs in which free ammonia is transferred from the wastewater into the gas phase (Li et al., 2018). Ammonia stripping experiments have been carried out by using a variety of techniques at different operating conditions such as pH, temperatures, and stripping agents (air, nitrogen, steam, or biogas) (Bonmati and Flotats, 2003; Georgiou et al., 2019; Huang et al., 2019; Jiang et al., 2013; Ledda et al., 2013; Li et al., 2018; Nie et al., 2015; Serna-Maza et al., 2015; Walker et al., 2011; Zeng et al., 2006). Successful applications of ammonia stripping have been reported both as a pre-AD process to various feedstocks, such as: domestic and industrial wastewaters, piggery and poultry wastes (Georgiou et al., 2019; Huang et al., 2019; Li et al., 2018; Zhang et al., 2012), and as a post-AD process to the effluent of anaerobic digestion and digested dairy manure supernatant (Huang et al., 2016; Jiang et al., 2014; Lei et al., 2007; Serna-Maza et al., 2015).

Currently, ammonia stripping methods are based on pH and/or temperature of substrate. Because pH directly affects the distribution of ammonia forms (NH_4^+/NH_3) in aqueous solutions, raising pH to more than 11 results in the conversion of almost all the ammonium ions (NH_4^+) into dissolved ammonia gases (NH_3) (Tchobanoglous et al., 2003).

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Table 1
Characteristics of chicken manure wastewater samples.

Parameter	Unit	Raw chicken manure wastewater	Chicken manure wastewater after air stripping
COD	(mg/L)	28,182 ± 2436	25,927 ± 2525
BOD_5	(mg/L)	2027 ± 204	1920 ± 150
TKN	(mg/L)	1270 ± 162	438 ± 18
NH_3-N	(mg/L)	1026 ± 140	212 ± 48
TS	(mg/L)	29,020 ± 1804	30,470 ± 1422
TVS	(mg/L)	16,095 ± 652	17,382 ± 1197
DO	(mg/L)	0.20 ± 0.03	2.56 ± 0.43
pH		7.27 ± 0.16	9.30 ± 0.21

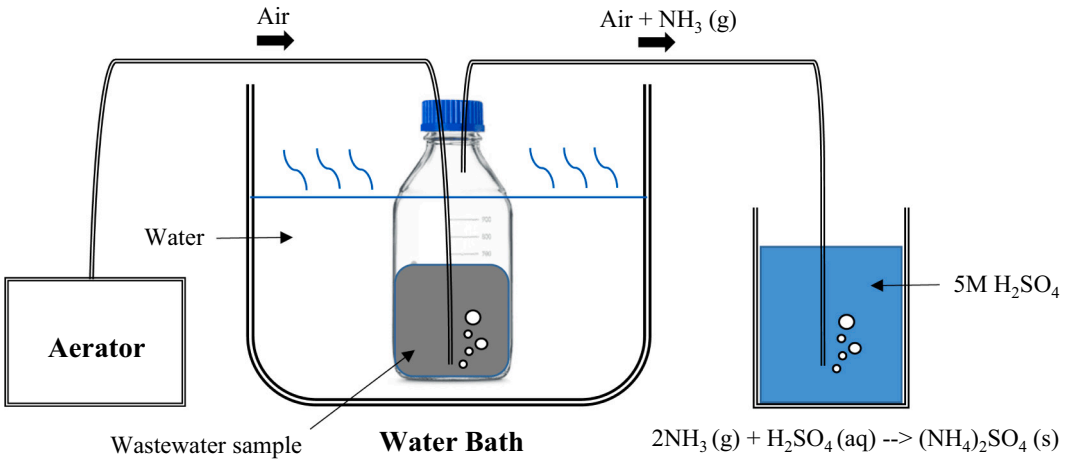


Fig. 1. Schematic of air stripping experimental setup.

Substrate pH has higher effect on NH_3 formation than temperature (Markou, 2015), but increasing pH can negatively influence the methanogenesis causing lower biogas production (Chen et al., 2008). Therefore, optimum conditions among pH, temperature and air flow rate for air stripping of ammonia should be investigated, especially when a cheap energy source is available.

The foremost objective of this study was to enhance biogas production from the AD of chicken manure wastewater by ammonia removal using air stripping as a pre-AD process. The specific objectives were: (1) to determine optimum conditions for ammonia removal by air stripping, including determination of reaction rate constants and design criteria, and (2) to determine the efficiency of biogas production from the AD using air stripping as a pre-AD process.

2. Methodology

2.1. Materials

Chicken manure wastewater samples were collected from a chicken farm located in Mae Chan District, Chiang Rai, northern Thailand. The main characteristics of the chicken manure wastewater are shown in Table 1. Digested sludge collected from the anaerobic lagoon of the Jampa farm wastewater treatment plant (Chiang Rai, Thailand) was used as an inoculum for bio-methane potential (BMP) test.

2.2. Air Stripping experiments

Air stripping experiments were conducted with a 1-L Duran glass bottle in a water bath (WNE22, Memmert, Germany) equipped with an aerator (ACO-9905, Hailea, China) and gas absorber, as illustrated in Fig. 1. The water bath was used to adjust temperatures of the experiments. The gas absorber containing 5 M H_2SO_4 solution was used to absorb the stripped ammonia gas to prevent releasing of harmful gas to the environment. Each experiment was performed with 600 mL of chicken manure wastewater sample, and the operating conditions were controlled at the initial pH values of 7, 9 and 11; temperatures of 30, 50, 70, 80, and 90 °C; and air flow rates of 0.5 and 4.0 L/min. The wastewater samples were taken for analysis of the remaining ammonia concentrations at various stripping times.

2.3. Bio-methane potential (BMP) Test

BMP tests of the chicken manure samples, with and without air stripping pre-treatment, were conducted in triplicate using 1 L lab-scale anaerobic digesters. Each testing digester was filled with 400 mL of the mixture of chicken manure sample and inoculum at the substrate/inoculum ratio of 0.5 gVS/gVS (Esposito et al., 2012). To create an anaerobic condition, all digesters were then filled with sufficient nitrogen gas to purge oxygen gas. Blank digesters, filled with 400 mL of the inoculum alone, were operated in parallel for data comparison. The BMP tests were carried out in a shaking water bath (WNE45, Memmert, Germany) controlled at the temperature of 37 °C, for 30 days. The produced biogas samples obtained from the BMP tests were collected and analyzed for their quantity and composition.

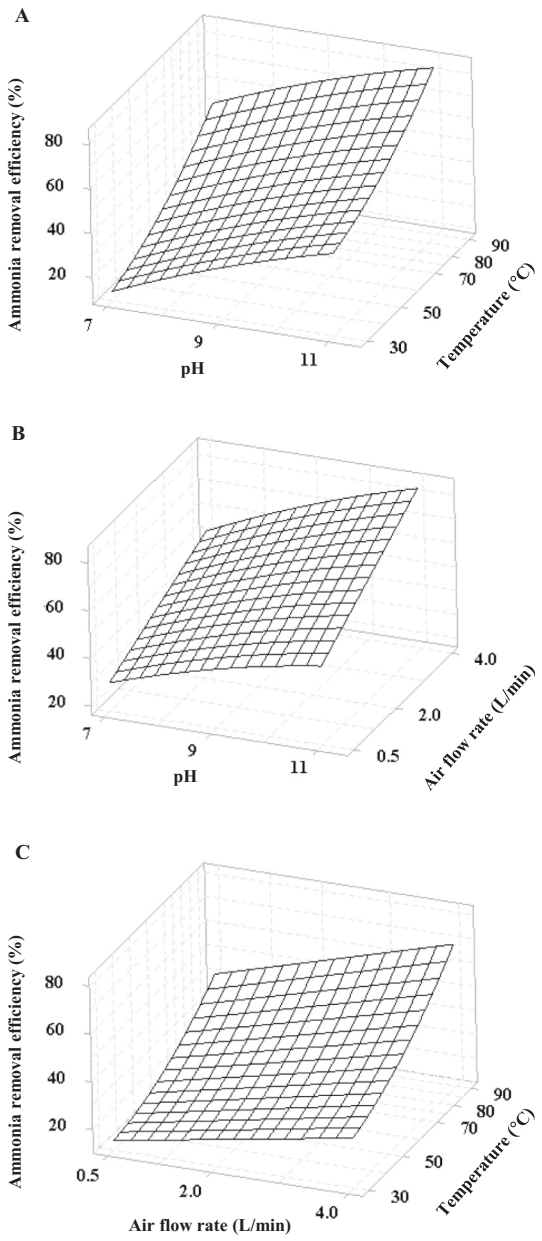


Fig. 2. Surface plots of ammonia removal efficiency versus air stripping conditions, A: pH - temperature, B: pH - air flow rate, and C: air flow rate – temperature, operating for 300 min.

2.4. Analytical methods

Characteristics of the chicken manure samples such as chemical oxygen demand (COD), biochemical oxygen demand (BOD_5), total solids (TS) and total volatile solids (TVS) were determined according to the Standard Methods no. 5220-C, 5210-B, 2540-B and 2540-E, respectively (APHA, 2012). Total Kjeldahl nitrogen (TKN) and ammonia nitrogen (NH_3−N) were measured using an automatic distillation unit machine (UDK149, VELP SCIENTIFICA, USA). Significant effects of the process parameters were statistically analyzed according to the response surface methodology (RSM) (Montgomery, 2005). Ammonia removal efficiency was calculated by using Eq. (1) as follows,

$$E = (c_0 - c)/c_0 \text{ x } 100 \quad (1)$$

where E is the ammonia removal efficiency (%), c_0 and c are the initial and remaining NH_3-N concentrations in the chicken manure wastewater (mg/L), respectively.

2.5. Kinetic model

The first-order kinetic model applied to air stripping process for ammonia removal is a simple model which can be expressed as

$$\frac{dc}{dt} = r = -kc \quad (2)$$

This equation can be integrated between c_0 and c, and $t = 0$ and t, to get the linear equation as follow:

$$ln\ c = ln\ c_0 - kt \quad (3)$$

where c is the concentration of NH_3-N remaining in the chicken manure (mg/L) at stripping time t (min), c_0 is the initial concentration of NH_3-N in the chicken manure (mg/L) at stripping time 0, and k is the reaction rate constant. To determine k, the concentrations of NH_3-N (ln$c$) versus stripping times (t) were plotted. A negative slope obtained from this linear graph is the k value.

The k value, which is temperature dependent, is commonly expressed in the Arrhenius equation.

$$k = Ae^{-\frac{Ea}{RT}} \quad (4)$$

$$lnk = lnA - \frac{E_a}{R}\frac{1}{T} \quad (5)$$

where T is the absolute temperature (K), A is the pre-exponential factor, Ea is the activation energy for the reaction (J/mol), R is the ideal gas constant (8.314 J/mol.K). The Ea value was determined by graphically plotting of ln$k$ versus 1/T. A negative slope obtained from this linear graph is the -Ea value.

3. Results and discussion

3.1. Effects of pH, temperature and air flow rate

Ammonia removal efficiencies obtained from the experiments at various air stripping conditions are shown in Fig. 2. A normal plot of the standardized effects (response is ammonia removal efficiency and α = 0.05) shown in Fig. 3 indicates that the initial pH, temperature and air flow rate had a significant effect on ammonia removal efficiency. The increases in the initial pH from 7 to 9 and 9 to 11 (Fig. 2A and B) caused the ammonia removal efficiencies to increase from 40 to 65% and from 65 to 80%, respectively, probably because most of the ammonia nitrogen was transformed to become free ammonia gas (NH_3), suitable for air stripping. At the pH levels above 9, the ammonia gas was predominant (Tchobanoglous et al., 2003). After completion of the air stripping process, due to the removal of free ammonia gas, the final pH values were decreased from 9 and 11 to about 8.5 and 9.7, respectively.

Fig. 2A and C revealed that high ammonia removal efficiencies were obtained at the elevated wastewater temperatures. At higher temperatures, the ammonia gas became more vapored and could be easier removed from the chicken manure samples with the supplied air. More than 60% and 80% of removal efficiencies were obtained at the

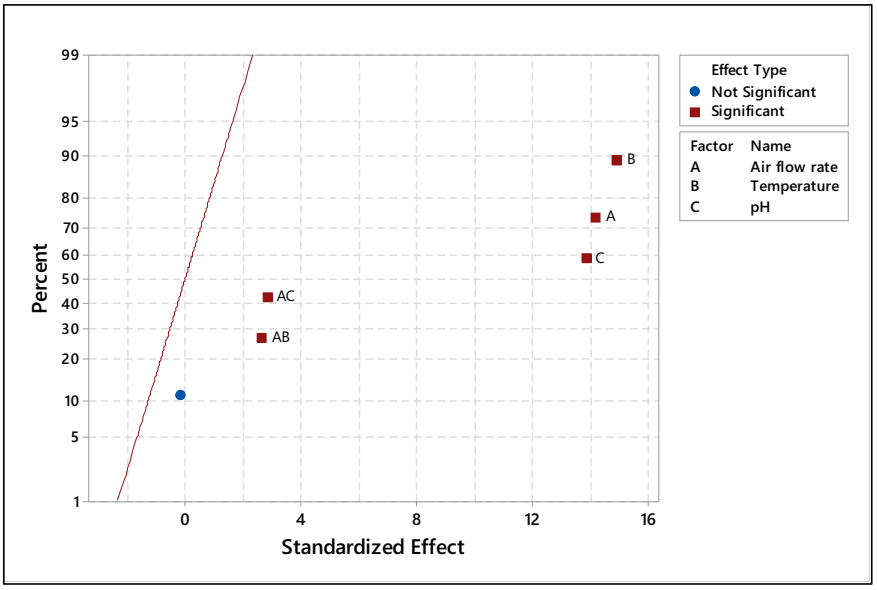


Fig. 3. Normal plot of the standardized effects relative to distribution fit line (response is ammonia removal efficiency and $\alpha = 0.05$).
Note: Distribution fit line is a line that presents the case when all the effects are zero. Positive effects (right side of distribution fit line) increase the response when the factors change from low to high value. Negative effects (left side of distribution fit line) decrease the response when the factors change from low to high value. The distance from the distribution fit line presents a statistically significant level. Effects further from zero are more statistically significant.

Table 2
Air stripping operating options to achieve ammonia removal efficiency of more than 60%.

Option	Initial pH	Temperature (°C)	Air flow rate (L/min)
1	≥9	90	≥2.8
2	≥9	≥85	4
3	≥10	90	≥2.2
4	≥10	≥70	4
5	11	≥80	≥2.2
6	11	≥65	4

temperatures higher than 65 and 80 °C, respectively. Similarly, the higher air flow rates also resulted in higher ammonia removal efficiencies in which more than 80% could be achieved at the air flow rate of 4.0 L/min.

The interactions among the initial pH, temperatures and air flow rates on ammonia removal efficiencies are represented in Figs. 2 and 3. The interactions between air flow rate – temperature (Fig. 3, factor AB) and air flow rate – initial pH (Fig. 3, factor AC) had significant effects on ammonia removal efficiency. Statistical analysis results in Fig. 3 indicated that the ammonia removal efficiency could be increased with increasing both air flow rate and temperature (or air flow rate and initial pH), simultaneously.

At the neutral pH of 7, increasing the air flow rates did not have significant effects on the ammonia removal efficiency. Even at the air flow rate of 4.0 L/min, the ammonia removal efficiencies were not higher than 60% (Fig. 2). At the initial pH of 11, operating at temperatures and air flow rates of more than 80 °C and 2.2 L/min, respectively, resulted in ammonia removal efficiencies of more than 60%.

At the lower temperatures (30 and 50 °C), increasing the air flow rates did not have significant effects on the ammonia removal efficiencies, but increasing initial pH resulted in increased ammonia removal efficiencies. When the air stripping was operated at the temperature of 90 °C, the ammonia removal efficiencies of more than 60% could be achieved at the operating air flow rates and initial pH of more than 2.8 L/min and 9, respectively.

If operating the air stripping process at the low air flow rate of 0.5 L/min, increasing both temperature and initial pH did not produce significant effects on the ammonia removal efficiencies. However, at the higher air flow rate of 4.0 L/min, increasing the temperatures and initial pH to more than 85 °C and 9, respectively, resulted in increased ammonia removal efficiencies of more than 60%.

The experimental results shown in Fig. 2 summarily indicated that the ammonia removal efficiencies of more than 80% could be achieved when the air stripping process was conducted at the initial pH, temperature and air flow rate of 11, 80 °C, and 4.0 L/min, respectively. The response optimization analyzed by using RSM suggested the options of the air stripping conditions to achieve the ammonia removal efficiencies of more than 60%, as shown in Table 2.

3.2. Kinetic models

Because the ammonia removal efficiencies were influenced by the stripping temperatures, the ammonia removal kinetics were determined at various temperatures, as shown in Fig. 4. The obtained kinetic models and k values are listed in Table 3, with coefficients of determination (R^2) being more than 0.90, which are comparable with those reported in the literatures. These experimental results indicated that ammonia removal reaction was a good fit for the first-order reaction, although some k values were different because of different stripping conditions employed in the previous studies (Serna-Maza et al., 2015; Walker et al., 2011). However, the k values obtained from this study should be applicable for the design and operation of the air stripping process for ammonia removal from chicken manure wastewater.

Estimation of temperature effects on reaction rates using the Arrhenius equation is shown in Fig. 5, with the R^2 value of 0.9545. The Ea and A values were found to be 18,150 J/mol and 1.232, respectively. The Arrhenius equation used to adjust the value of the reaction rate constant, k, to reflect the effect of temperature can be expressed in Eq.

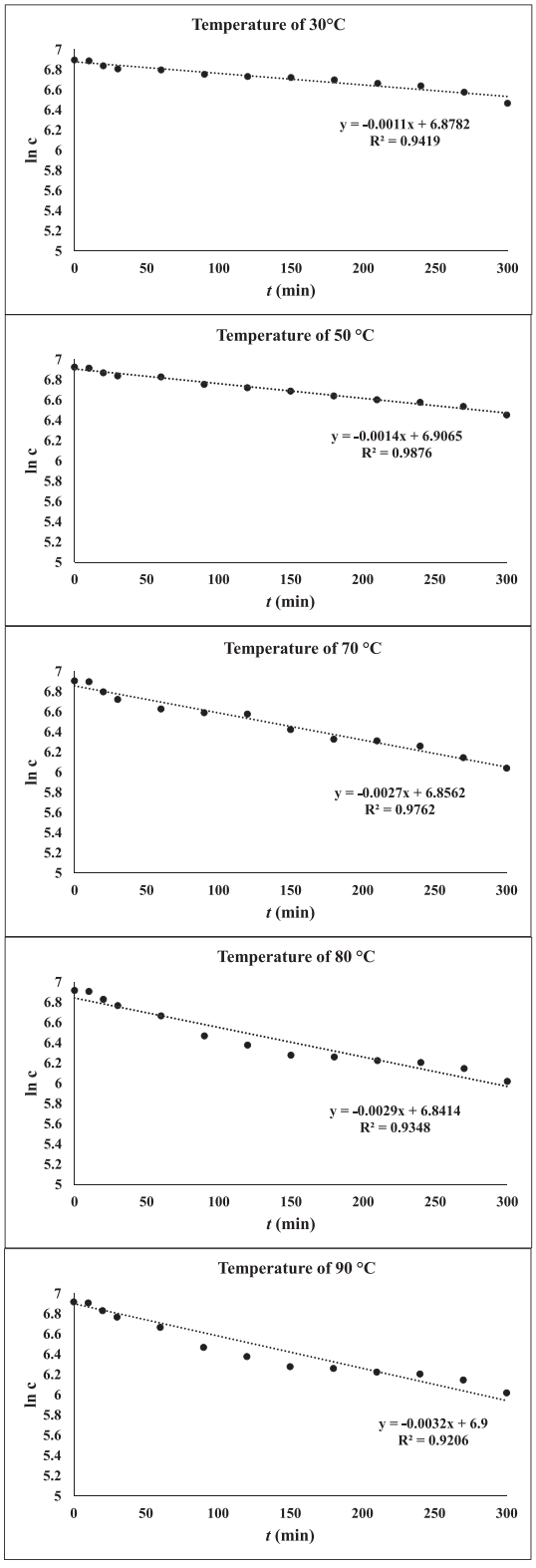


Fig. 4. Ammonia concentrations with stripping times at temperatures of 30, 50, 70, 80, and 90 °C, initial pH of 9 and air flow rate of 4.0 L/min.

Table 3
Kinetic models of ammonia removal by using air stripping process at various temperatures, operating at initial pH of 9 and air flow rate of 4.0 L/min.

Temperature (°C)	Kinetic model	k	R^2	k (previous studies)
30	$\ln c = \ln c_0 - 0.0011(t)$	0.0011	0.9419	0.0011–0.0014[a]
50	$\ln c = \ln c_0 - 0.0014(t)$	0.0014	0.9876	0.0053–0.0068[a]
70	$\ln c = \ln c_0 - 0.0027(t)$	0.0027	0.9762	0.0162–0.0267[a] and 0.031[b]
80	$\ln c = \ln c_0 - 0.0029(t)$	0.0029	0.9348	
90	$\ln c = \ln c_0 - 0.0032(t)$	0.0032	0.9206	

[a] Results of Serna-Maza et al. (2015): stripping conditions; using biogas as a stripping agent, temperatures of 35, 55 and 70 °C, air flow rate of 0.25 L/min, pH 10.
[b] Results of Walker et al. (2011): stripping conditions; using biogas as a stripping agent, temperatures of 70 °C, air flow rate of 0.125 L/min, no pH adjustment.

(6).

$$ln\ k = 0.209–2183(1/T) \quad (6)$$

3.3. Biogas Production

Although air is frequently employed as a strip gas in ammonia stripping process (Huang and Shang, 2006), oxygen contained in the air could act as a strong inhibitory agent to the anaerobic microorganisms such as acetogens and methanogens involved in the methane production (Whitman et al., 2006). From the results shown in Table 1, after the air stripping process, the COD, BOD_5, TS and TVS concentrations were found to be slightly changed, but the NH_3-N concentrations were significantly decreased while DO concentrations were increased.

To evaluate the effects of the air stripping process on methane production, the BMP tests were conducted. Fig. 6 shows the cumulative biogas production from the chicken manure with and without air stripping pre-treatment process during the 30 days operation. The chicken manure sample after the air stripping process exhibited slower biogas production, but eventually resulted in higher biogas yield than that without air stripping. The biogas yields of about 57.3 and 45.2 L per kg-VS were achieved from the AD of the chicken manure wastewater samples with and without air stripping pre-treatment, respectively. These results revealed that, probably due to less effects of ammonia inhibition (Table 1), the air stripping pre-treatment of the chicken manure resulted in about 25% increase in biogas production.

4. Conclusions

The experimental results of this study suggested that the air stripping conditions such as initial pH, temperature, and air flow rate had significant effects on the ammonia removal efficiencies. To achieve the ammonia removal efficiency of 80%, the optimum operating conditions for the initial pH, temperature, and air flow rate were found to be 11, 80 °C, and 4.0 L/min, respectively. The ammonia removal kinetics were found to fit with the first-order model. By using the air stripping process as a pre-treatment of the AD, the biogas production was found to increase by about 25%.

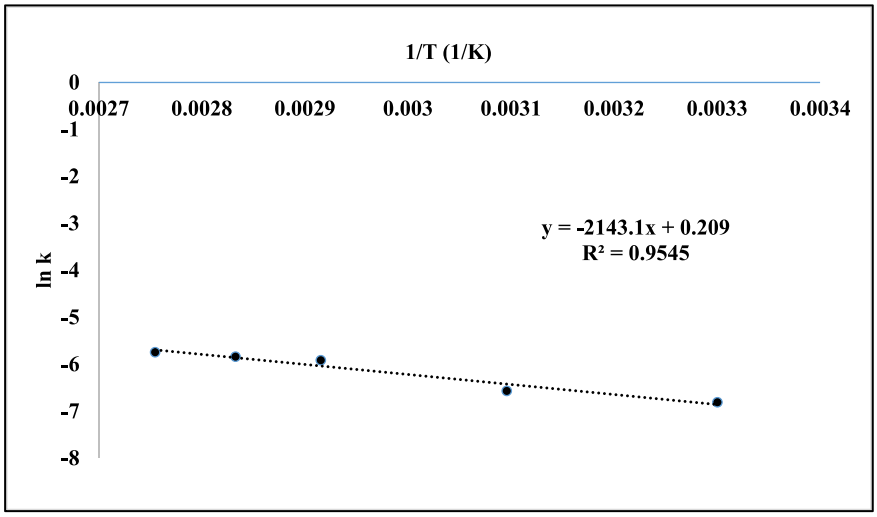


Fig. 5. Plot of reaction rate constants (ln*k*) with temperatures (1/*T*).

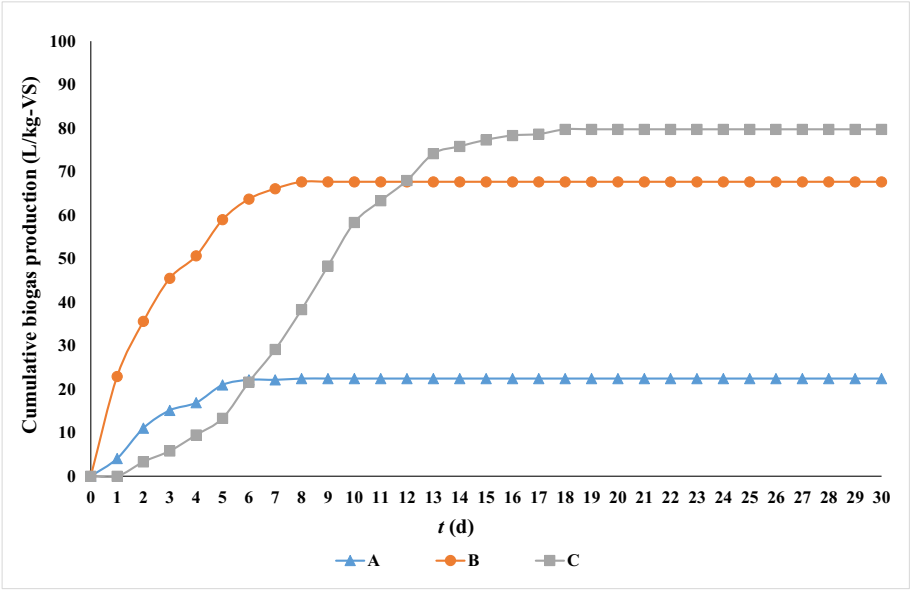


Fig. 6. Cumulative biogas productions from BMP tests.
A: Inoculum.
B: Chicken manure (without air stripping) + inoculum.
C: Chicken manure (with air stripping) + inoculum.

CRediT authorship contribution statement

Krailak Fakkaew: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing – original draft. **Chongrak Polprasert:** Conceptualization, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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